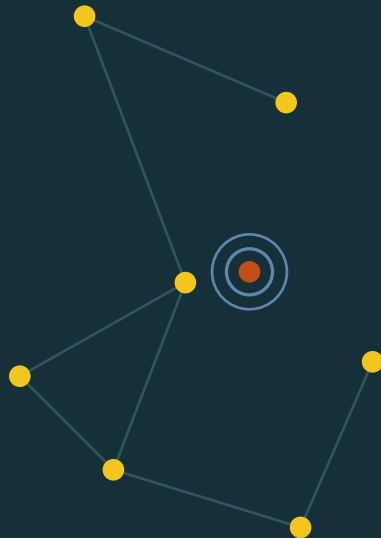


NSF REU · COMPUTATIONAL MODELING SERVING PORTLAND

Placing Clean-Air Shelters for Wildfire Smoke

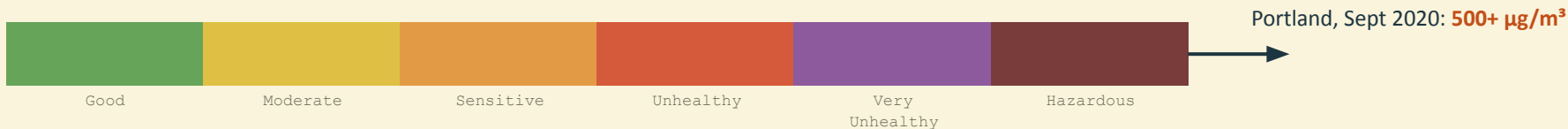
A vulnerability-aware agent-based model of how wildfire smoke reaches Portland's unsheltered residents — and who it harms most.

Fatima Asghar · Mentor: Dr. Christopher Teuscher, Portland State University



Unequal harm from the same air

Wildfire smoke is worsening across the West — and the same air does not do the same harm to everyone breathing it.

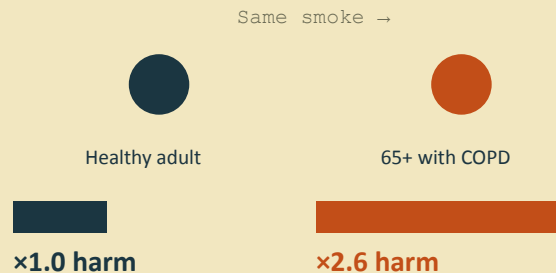


Worst urban air on Earth that year — with nowhere indoors to retreat if you are unsheltered.

The blind spot

Screening tools like EJScreen assume identical air means identical harm.

Environmental epidemiology has shown for over a decade that it doesn't: age and chronic lung disease can multiply the damage from the very same concentration of PM2.5.



Siting for vulnerability, not just exposure

Where should a city place clean-air shelters to protect its unsheltered residents — and does the answer change when placement accounts for who is most *biologically vulnerable*, not just who is most *exposed*?

1

A planning tool, not a post-mortem

Prospective by design — built to guide siting for the next smoke event, not just to grade how we handled the last one.

2

Grounded in a real event

Calibrated on Portland's Sept 7–19, 2020 smoke episode, one of the worst air-quality stretches on record.

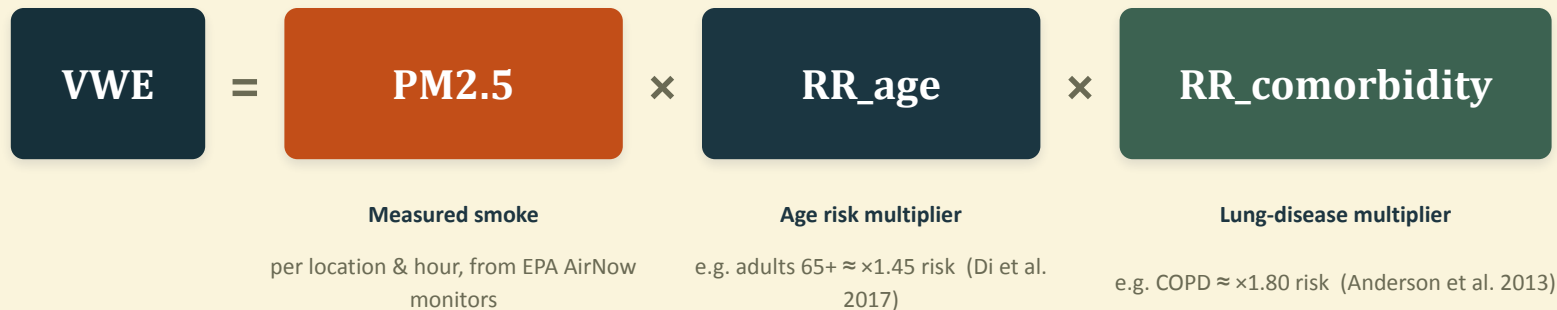
3

Open & reusable

Reproducible, MIT-licensed code that any city can point at its own streets, shelters, and population.

A vulnerability-weighted exposure metric

One number per person, per hour — the project's central metric.



Why it matters: the highest-smoke places and the highest-vulnerability places aren't the same places — so the "best" shelter map can move.

Five placement strategies

Run each strategy through the same simulation and compare the outcomes.

01

Status quo

Today's shelter map — the baseline everything else is measured against.

02

Density

Put new shelters where the most unsheltered people already cluster.

03

Gap-index

Target populous areas that current shelters leave underserved.

04

► BENCHMARK

PM2.5 oracle

Full-information ideal: relieve the most raw smoke exposure.

05

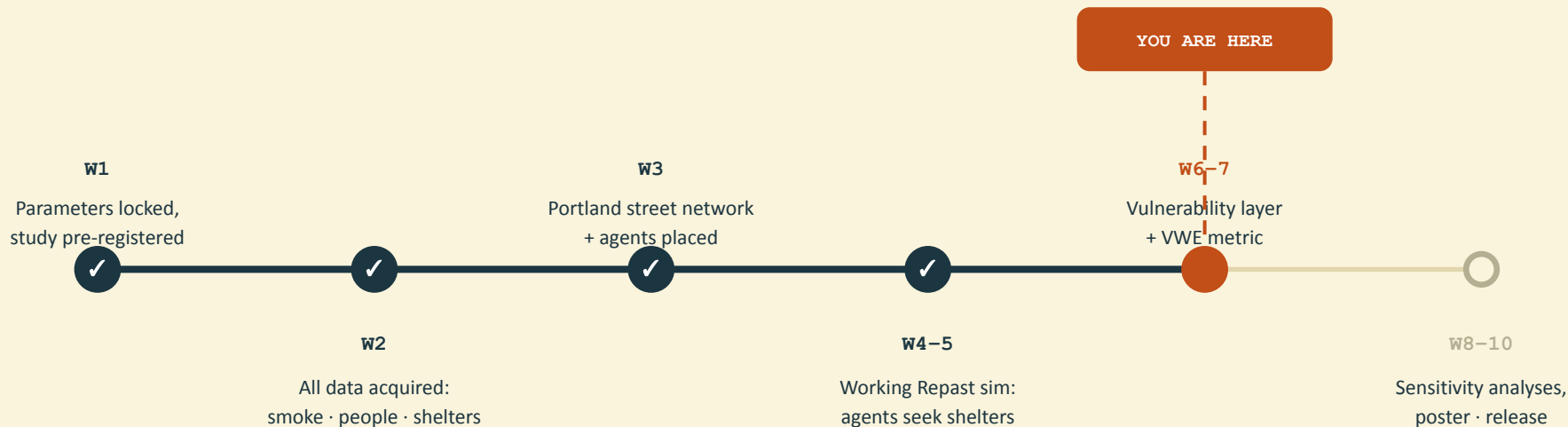
► BENCHMARK

VWE oracle

Full-information ideal: relieve the most vulnerability-weighted smoke.

Scored on: exposure-hours above the "unhealthy" line · **Gini coefficient** (how equally exposure is shared) · swept across mobility & comorbidity assumptions.

Where the work stands



Headline: the simulation already runs end-to-end in Repast — synthetic residents walk the real Portland street grid toward the nearest shelter they can reach, and I can total up each person's smoke dose over the whole event.

Implementation in Repast Symphony

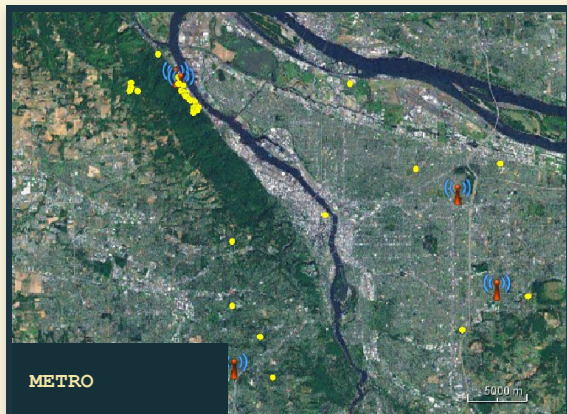
Started from Repast's stock GIS **Geography** demo → stripped most of it → kept and repurposed **three** components.



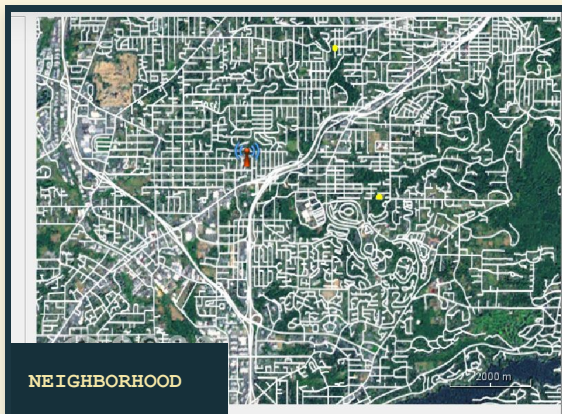
Agent behavior: walk along connecting streets toward the nearest shelter you can actually reach.

The simulation in action

unsheltered agent



≈ 5 km across — shelters span the whole region



≈ 2 km — agents thread the street grid



≈ 1 km — agents cluster toward a shelter

Same model, three zoom levels. Every buttercream dot is a simulated resident moving on the real Portland network; every tower is a clean-air shelter they can walk to.

The road to August

Finish the model

- Wire the age- and comorbidity-based vulnerability layer into every agent, then build the VWE oracle strategy
- Run the full parameter sweep — five strategies crossed with mobility and comorbidity assumptions

Make it count

- Run RR and comorbidity-prevalence sensitivity analyses across the published confidence intervals
- If time allows, run a BenMAP cost-effectiveness pass — cases and dollars averted per strategy
- Publish the community brief and poster, and release the code with a permanent Zenodo DOI

→ **By August 14:** *a working model, a pre-registered study, and an open, reproducible tool for future smoke events.*